

1 atavide-lite: simplified atavistic metagenome 2 processing

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10 Summary

11 atavide lite is a modular metagenomics processing pipeline designed to handle complex,
12 multi-sample, multi-technology datasets. Instead of a single monolithic workflow, it separates
13 processing into discrete, independent steps, allowing users to run only what they need, retry
14 failed components without restarting the entire workflow, and optimise resource use for their
15 specific environment. The pipeline integrates both read-based and assembly-based approaches,
16 and supports both paired-end short reads and long-read sequencing data.

17 Statement of need

18 All environments are impacted by microbes and viruses. Over the last two decades great
19 strides have been made in exploring what microbes and viruses are present in a wide swath
20 of environments, and understanding what they are doing, and how they contribute to
21 biogeochemical cycles, health, or disease.

22 Driven by decreases in the costs of DNA sequencing, new DNA sequencing technologies, and
23 the availability of abundant and cheap computing, metagenomics has emerged as one of the
24 most complete methods to understand the microbes and viruses in any environment. Typically,
25 metagenomics processing starts by size partitioning the microbial components to focus on
26 the bacteria or viruses, and then extracting all the DNA present in the sample. Advanced
27 computational techniques are then applied to explore which microbes and viruses are present
28 (their taxonomy) and the functions that they are performing (by annotating their genes). By
29 leveraging the additional information that multiple samples provide, metagenomic binning
30 strives to reconstruct complete, or near-complete, microbial and viral genomes from complex
31 samples.

32 atavide_lite performs all the computational processing of metagenomics sequence data,
33 starting with raw reads from a DNA sequencing machine, we end up with annotated reads and
34 reassembled genomes.

35 State of the field

36 Numerous metagenomics pipelines exist, from web-based systems like MG-RAST ([Meyer et
37 al., 2008](#)), to a proliferation of command-line pipelines ([Blanco-Míguez et al., 2023](#); [Garfias-
38 Gallegos et al., 2022](#); [Laudadio et al., 2019](#); [Lu et al., 2022](#); [Roach et al., 2024](#); [Tyagi &](#)

39 [Katara, 2024](#); [Walker et al., 2022](#); e.g. ?). Workflow management systems like Nextflow ([Di](#)
40 [Tommaso et al., 2017](#)) and Snakemake ([Köster & Rahmann, 2012](#)) streamlined the creation of
41 pipelines for bioinformatics analysis, including metagenomics.

42 Most metagenomics pipelines work process sequence data in the same way. Each starts with
43 quality control of the raw *fastq* files retrieved from the DNA sequencing facility. Quality
44 control includes removing low quality reads, removing adapters and linkers that remain in the
45 sequences, and removing overly short or long reads. Next, if the sample is a host-associated
46 sample, the pipelines remove reads that map to the host genome. At this point, the pipelines
47 often branch into read-based analysis and contig-based analysis. For read-based analysis,
48 individual reads are annotated for taxonomy and function. The number of reads that map
49 to each category is a proxy for the abundance of that category in the original sample, and so
50 mapped reads are counted and their abundance normalised. For contig-based analysis, the
51 contigs are grouped or binned into sets of contigs that likely came from the same organism,
52 and then those contigs are annotated.

53 The continued decrease in the cost of DNA sequencing, together with the ever increasing
54 sequencing throughput and computing power, mean that these days we routinely generate
55 metagenomics datasets with hundreds or thousands of samples, and each sample needs to be
56 processed through this pipeline.

57 Managing this many samples across real-world, large-scale deployments reveal two persistent
58 problems:

- 59 1. Fragility at scale When processing thousands of datasets, there are multiple failure points
60 in each metagenomics analysis.
- 61 2. Poor portability across HPC environments: Optimising for one cluster often breaks
62 performance or compatibility on another, especially when each platform has idiosyncratic
63 storage, job scheduling, and runtime constraints.

64 For example, when processing thousands of samples from the Sequence Read Archive (SRA)
65 ([Leinonen et al., 2011](#)), we found that some samples failed to download, and we had to
66 interrupt the pipeline to complete the downloads, fix the issue, and restart the pipeline.
67 When processing our own data, corrupt or inconsistent sequence files (e.g. *fastq* files with
68 different length sequences or quality scores), can crash a run. Similarly, when using large
69 computations, such as comparing sequence reads to a database of reference sequences using
70 MMSeqs ([Steinegger & Söding, 2017](#)), the computation occasionally times out because of
71 limitations of the compute environment, the compute fails because of unforeseen software
72 incompatibility, or the computation fails for a variety of other reasons. Although pipelines
73 streamline and seemingly simplify the process, our experience showed that complex pipelines
74 fail with different samples at different states, obfuscating the precise cause of the failure.

75 Moreover, each of the computational platforms we use has nuances for the most efficient
76 computational processing. For example, our institutional HPC has a very fast local disk which is
77 available from a non-standard location and is only available to compute nodes. The Australian
78 National Computational Infrastructure's Gadi system does not support large array jobs. The
79 Pawsey Supercomputing Centre's Setonix system regularly deletes files on their "/scratch"
80 disk, but has a large S3 storage system that is available to all compute nodes but from a
81 specific name.

82 *atavide_lite* addresses these challenges by replacing an "all-or-nothing" execution model of
83 most computational pipelines with independent, platform-optimised steps. We provide generic
84 scripts that will run on most clusters, plus tuned configurations for the HPC systems we use,
85 and encourage users to adapt them to their own environments. We provide an issue form that
86 allows users to request configurations for specific systems, and provide documentation to enable
87 agentic-AI development of the pipelines tailored to any system. This design improves fault-
88 tolerance, eases debugging, and enables efficient use of heterogeneous compute infrastructures.

Software design

The overall atavide_pipeline is presented in [Figure 1](#).

Design decisions

When designing atavide_lite we have attempted to make code that is re-usable on any system, and to simplify the processing steps. Specific design decisions we took include:

User defined variables when possible.

We use a generic DEFINITIONS.sh file to define some common variables, and that should reside in each analysis directory and becomes part of the analysis.

Similar analysis for single-end and paired-end reads.

Although different instruments and technologies vary drastically, once the sequences themselves are generated, the analysis is usually quotidian. Wherever possible, we use the same analysis for all sequence read data.

Using array jobs when possible.

Array scheduling on HPC systems allows users and administrators to issue a single command to control many tens or hundreds of jobs. Each cluster has different limits on how many jobs may be a part of an array, and this will be determined by your systems administrators, but generally more than 100 jobs are permissible in a single array.

Leverage HPC scheduler controls when possible.

Most HPC systems have a controller that can start queued jobs based on dependencies. When possible, we queue all of the jobs at once, but allow the controller to start the dependencies if, and only if, the preceding job successfully completed. Depending on the specific HPC setup, dependent jobs may be marked as unable to complete or removed from the queue. If a job fails, the output and the logs will be clear where the failure point occurred so that the specific failure issue can be fixed before the computation is restarted.

Computational steps in the pipeline.

Before the data is processed, there are two steps that the user needs to perform. First, we create a DEFINITIONS.sh file in the working directory. This file defines a few variables that are used throughout the processing. First, we define the file ending for the fastq and fasta files. These endings are typically sequence instrument vendor specific (e.g. MGI-generated fastq files may have different file endings from Illumina-generated fastq files), and are trimmed from later files. We define a sample name which covers the whole dataset, and we define the location of the host genome sequence and where to save the host-related and not host-related sequences. Note the last three can be omitted if host mapping is not required.

Next, we create an R1_reads.txt or reads.txt file and either a NUM_R1_READS or NUM_READS variable. The R1_reads.txt file is required for processing paired end reads, while the single end reads uses reads.txt. The variables are really just a simple convenience and are not strictly necessary for processing the jobs, but make it easier to submit array jobs.

We start with quality control using *fastp* ([Chen et al., 2018](#)) to remove any sequence with too many N bases, remove adapters, and reads that are too short. For ONT data, our defaults are two N's and a minimum length of 50 bp, while for short-read data, our defaults are 1 N base and a minimum length of 100 bp.

The second step, if required, is to remove host DNA. `atavide_lite` maps the reads that pass quality control to the host genome using *minimap2* (H. Li, 2018), and then uses *samtools* (H. Li et al., 2009) to filter the reads. Sequences that do not match the flag 3588 – and thus represent mapped reads which pass the platform/vendor quality check, are not PCR or optical duplicates, and are primary alignments – are filtered as mapping to the host genome. For short reads, additional *samtools* flags are used to identify R1 reads (matching flag 65) and R2 reads (matching flag 129). Reads that do not match the flag 3584 – and thus represent unmapped reads which pass the platform/vendor quality check, are not PCR or optical duplicates, and are primary alignments – are filtered as unmapped reads (H. Li et al., 2009) and again, short reads were separated using flags that match 77 for R1 reads and 141 for R2 reads. The matching and unmapping reads are stored separately, although currently we do not use the reads that match the host genome for any downstream analysis.

Note that the host genome can be any genome or multi-fasta record that the user wants excluded from subsequent analysis, and is defined in the aforementioned `DEFINITIONS.sh` file.

Reads that do not map to the host genome are converted to fasta format, and then mapped against the UniRef100 database (Suzek et al., 2007) using the easy-taxonomy workflow built into `MMseqs2`.

`MMseqs2` easy-taxonomy parameters were optimised by the authors for a balance of sensitivity and speed and include a 7-mer double-match prefilter with compositional bias correction, low-complexity masking with TANTAN, and default similarity thresholds ($-k$ -score ~ 95) that generate sufficient similar k -mers for accurate detection while maintaining computational efficiency. Candidate hits are further filtered by fast ungapped alignment and refined with vectorized Smith-Waterman alignments, after which taxonomic labels are assigned using a lowest common ancestor (LCA) approach described in more detail in their paper and on their website (Steinegger & Söding, 2017). The taxonomies are reformatted with *taxonkit* (Shen & Ren, 2021) and merged into a single table.

The functional annotations were enriched by mapping the UniProt IDs from the `MMseq2` output directly to proteins associated with BV-BRC subsystems (Olson et al., 2023) (this was a 1:1 mapping and did not require thresholds/cutoffs). The number of reads that `MMseqs2` reported mapped to each protein was counted, and the total was divided by the number of mapped reads to normalise the read counts.

For the assembly-based approaches, the metagenomes were assembled using *megahit* (D. Li et al., 2015) which by default employs a succinct *de Bruijn* graph approach with multiple k -mer sizes (21 to 141 in steps of 20), automatic memory optimization, and conservative heuristics for contig pruning to balance assembly sensitivity, contiguity, and computational efficiency. Metagenome-assembled genomes (MAGs) were reconstructed using *VAMB* (Nissen et al., 2021), a variational autoencoder-based approach that jointly encodes k -mer composition and differential coverage profiles of contigs into a low-dimensional latent space. By learning these compressed representations, *VAMB* effectively separates contigs originating from different microbial genomes and clusters them using a medoid-based algorithm. This deep learning approach has been shown to improve bin purity and completeness compared to conventional binning methods such as *MetaBAT* (Kang et al., 2019) or *CONCOCT* (Alneberg et al., 2014), particularly in complex metagenomic datasets (Papudeshi et al., 2017).

Research impact statement

We routinely use `atavide_lite` to process metagenomics datasets from a wide range of environments and hosts. For example, we studied how microbes migrate through complex environments (Grigson et al., 2026) which did not require any host genome removal steps. In contrast, our studies on the microbial and viral communities of the CF airways [Carlson-Jones et al. (2026); Goddard2026-ee] and the causes of an undiagnosed illness in Central Africa (Wawina-Bokalanga et al., 2026) both required the removal of human genome contamination.

180 Meanwhile, when we studied the microbes on the surface of elasmobranchs (Doane et al., 2026)
181 we eliminated known shark genomes from our analysis. In each of these very different sampling
182 schemes, sequenced with different technologies, we used the same underlying pipeline.

183 AI usage disclosure

184 We widely use agentic AI in code development. Our typical development pipeline now includes
185 initial project scoping with ChatGPT leading to detailed instructions to write code that are
186 provided to either Codex or Claude Code. All commits and pull requests are reviewed with
187 Codex and Copilot in GitHub, and agentic AI is used to resolve issues. We provide detailed
188 documentation in our GitHub repository for other users to extend atavide_lite to new clusters,
189 and encourage them to use agentic AI in doing so.

190 However, this manuscript was written without AI support.

191 Figures

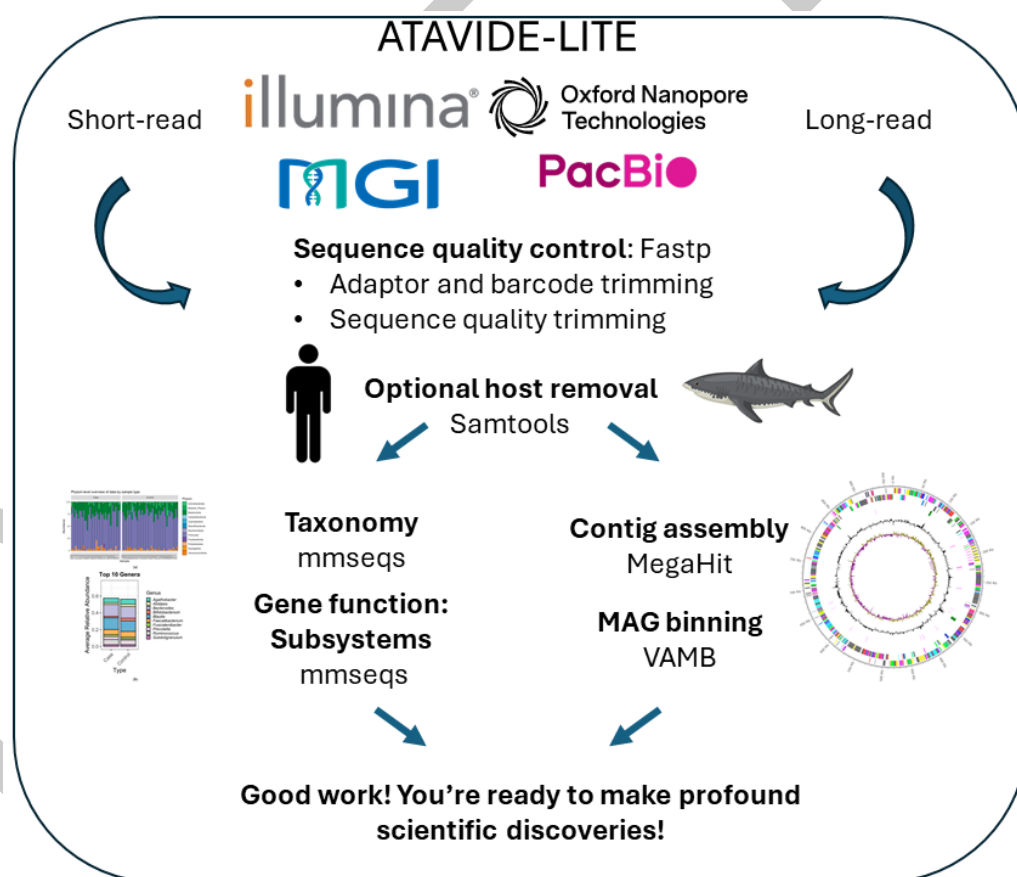


Figure 1: atavide_lite schematic by M. Doane demonstrating that we start with any type of sequence data, and process them through a similar pipeline for QC, host removal, annotation, assembly, and analysis.

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All computations were performed on Flinders HPC Deepthought (Flinders University, 2021) or Pawsey's HPC, Setonix (Pawsey Supercomputing Research Centre, 2023)

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